CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

1.2 SI Units

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (85 Marks)

Question 1 Easy [10 marks]

Part (a) [4 marks]

Table 1.1 shows some SI base quantities and their corresponding unit.

Table 1.1

quantity	SI base unit
mass	kg
time	
length	

(i) Complete Table 1.1 to show the SI base unit of each quantity.

[2]

(ii) State two other SI base quantities and their corresponding units.

[2]

Mark Scheme (4 marks):

(i) The SI base units are:

SI base quantity	SI base unit
mass	kg
time	***s***
length	***m***

- time = second **OR** s *[1 mark]*
- length = metre **OR** m *[1 mark]*

(ii) Two other SI base quantities & their units are:

Any **two** from:

- Temperature **AND** Kelvin / K *[1 mark]*

- Current **AND** amp / ampere / A *[1 mark]*
- Amount of substance **AND** mole / mol *[1 mark]*

[Total: 4 marks]

Part (b) [2 marks]

Newton's second law of motion describes the relationship between force *F* mass *m* and acceleration *a*:

$$F = ma$$

The derived unit for force is the newton (N).

Use Newton's second law to write 1 N in SI units.

Mark Scheme (2 marks):

Write 1 N in SI base units:

- $F = ma \quad \Rightarrow \quad F = ma$
  - m = kg
  - a = m s^{-2}
- 1 N = (kg) × (m s^{-2}) *[1 mark]*
- 1 N = kg m s^{-2} *[1 mark]*

[Total: 2 marks]

It is really important that you learn the SI base units in case you are asked a question like this in the exam

Once you know the SI base units, it is fairly straightforward to deduce the base units for any derived unit because the method is exactly the same as the method given here.

Part (c) [2 marks]

The kinetic energy equation describes the relationship between energy *E*, mass, *m*, and velocity, *v*.

$$E_k = \tfrac{1}{2}mv^2$$

The derived unit for energy is the joule (J).

Use the equation linking kinetic energy and speed to write 1 J in SI base units.

Mark Scheme (2 marks):

Write 1 J in SI base units:

- $E_k = \tfrac{1}{2}mv^2 \quad \Rightarrow \quad E_k = mv^2$
  - m = kg

- v = m s^{-1}

- 1 J = (kg) × (m s^{-1})^2 *[1 mark]*

- 1 J = kg m^2 s^{-2} *[1 mark]*

[Total: 2 marks]

The same method is used to determine the SI base units for any derived unit

Practice this skill as part of your revision. It will help you to remember the SI base units (if you don't already know them) rather than having to learn them by rote.

Part (d) **[2 marks]**

The derived unit for power *P* is the watt W.

Given that power is the energy transferred *E* per unit time *t*, deduce the SI base units for power

Mark Scheme (2 marks):

Deduce the SI base unit for W:

- Power: $P = \frac{E}{t}$ $\Rightarrow$ $P = \frac{E}{t}$
  - E = kg m^2 s^{-2} (from **(c)**)
  - t = s
- W = $\frac{kg \ \ m^2 \ \ s^{-2}}{s}$ *[1 mark]*
- W = kg m^2 s^{-3} *[1 mark]*

[Total: 2 marks]

It would be a good idea to practice deducing the SI base units for a variety of derived units so that you get comfortable with the method and familiar with the SI units

Try it for each topic as part of your revision.

Question 2 Easy [14 marks]

Part (a) [3 marks]

In the following list, underline all the units that are SI base units

kilogram newton metre coulomb volt ampere

Mark Scheme (3 marks):

Underline all the units that are SI base units:

<u>kilogram</u> newton <u>metre</u> coulomb volt <u>ampere</u>

- kilogram *[1 mark]*
- metre *[1 mark]*
- ampere *[1 mark]*

[Total: 3 marks]

Make sure you can quickly recall the SI quantities! This will save you a lot of time in your exam.

Part (b) [3 marks]

Complete Table 1.1 to show the name and symbol of each prefix and the corresponding power of ten.

Table 1.1

prefix	power of ten
tera (T)	
	10^{9}
	10^{3}
centi (c)	10^{-2}
milli (m)	
micro (μ)	

	10^{-9}
pico (p)	10^{-12}

Mark Scheme (3 marks):

The correct table of prefixes and powers of ten is as follows:

Prefix	Power of ten
tera (T)	***10^{12}***
giga (G)	10^{9}
kilo (k)	10^{3}
centi (c)	10^{-2}
milli (m)	***10^{-3}***
micro (μ)	***10^{-6}***
nano (n)	10^{-9}
pico (p)	10^{-12}

- *[3 marks]* for all six correct
- *[2 marks]* for four or five correct
- *[1 mark]* for two or three correct
- *[0 marks]* for less than two correct

[Total: 3 marks]

Part (c) **[4 marks]**

Convert the following measurements into standard form and to the specified number of significant figures.

i) 0.71 GJ to 3 s. f.

[1]

ii) 24.5 kW to 4 s .f.

[1]

iii) 0.003823 mm to 1 s. f.

[1]

iv) 644.9 pN to 2 s.f.

[1]

Mark Scheme (4 marks):

(i) 0.71 GJ in standard form and 3 s.f. is:

- 1 GJ = 1×10^9 J
- 0.71 GJ = $0.71 \times (1 \times 10^9$ J)
- 0.71 GJ = 7.10×10^8 J (3 s.f.) *[1 mark]*

(ii) 24.5 kW in standard form and 4 s.f. is:

- 1 kW = 1×10^3 W
- 24.5 kW = $24.5 \times (1 \times 10^3$ W)
- 24.5 kW = 2.450×10^4 W (4 s.f.) *[1 mark]*

(iii) 0.003823 mm in standard form and 1 s.f. is:

- 1 mm = 1×10^{-3} m
- 0.003823 mm = $0.003823 \times (1 \times 10^{-3}$ m)
- 0.003823 mm = 4×10^{-6} m (1 s.f) *[1 mark]*

(iv) 644.9 pN in standard form and 2 s.f. is:

- 1 pN = 1×10^{-12} N
- 644.9 pN = $644.9 \times (1 \times 10^{-12}$ N)
- 644.9 pN = 6.4×10^{-10} N (2 s.f.) *[1 mark]*

[Total: 4 marks]

You should be familiar with using scientific notation from IGCSE, but it can be tricky when you are first learning. Practice makes perfect.

Make sure you are confident in using and converting numbers in standard form well before your exams. And make sure you know how to quote numbers to appropriate numbers of significant figures, as this is something that can cost you marks in an exam if you don't do it correctly!

Part (d) **[4 marks]**

Convert the following from their standard form into measurements with the most suitable prefixes

9.3×10^{6} J = ... J

7.4×10^{3} W = ... W

1.0×10^{-2} m = ... m

5.3×10^{-9} C = ... C

Mark Scheme (4 marks):

The measurements with suitable prefixes are:

- 9.3×10^{6} J = 9.3 MJ *[1 mark]*
- 7.4×10^{3} W = 7.4 kW *[1 mark]*
- 1.0×10^{-2} m = 1.0 cm **OR** 10 mm *[1 mark]*
- 5.3×10^{-9} C = 5.3 nC *[1 mark]*

[Total: 4 marks]

Learn the prefixes and their magnitudes. This can give you easy marks in your exam.

Question 3 Easy [11 marks]

Part (a) [5 marks]

Scientists around the world use the same internationally agreed system of units. These are called the fundamental SI base units and are listed in Fig. 1.1.

Quantity	Name	Symbol
length		
	kilogram	
		s
	ampere	
temperature		K

Fig. 1.1

Complete the missing quantities, names and symbols in Fig. 1.1.

Mark Scheme (5 marks):

A correctly completed table that would score 5 marks should look like this:

One mark for each correct row:

Quantity	Name	Symbol	
length	***metre***	***m***	***[1 mark]***
mass	kilogram	***kg***	***[1 mark]***
time	***second***	s	***[1 mark]***
current	ampere	***A***	***[1 mark]***

temperature	***kelvin***	K	***[1 mark]***

[Total: 5 marks]

Part (b) [2 marks]

Use the equation linking kinetic energy and speed to write 1 Joule in SI base units.

Mark Scheme (2 marks):

Using the equation linking kinetic energy and speed to write 1 Joule in SI base units:

- Kinetic energy [J] = mass [kg] × (speed)^2 [m s^{-1}]^2 *[1 mark]*
- 1 joule = kg m^2 s^{-2} *[1 mark]*

[Total: 2 marks]

Part (c) [2 marks]

Use the equation *Charge* = *Current* × *Time* to write 1 Coulomb in SI base units.

Mark Scheme (2 marks):

Using the equation *Charge* = *Current* × *Time* to write 1 Coulomb in SI base units:

- Charge [C] = Current [A] × Time [s] *[1 mark]*
- 1 coulomb = 1 A s *[1 mark]*

[Total: 2 marks]

Part (d) [2 marks]

Use the equation *Energy* = *Charge* × *Potential Difference* to write 1 volt in SI base units.

Mark Scheme (2 marks):

Using the equation *Energy* = *Charge* × *Potential Difference* to write 1 volt in SI base units:

- Energy [J] = Charge [C] × Potential Difference [V] *[1 mark]*
- V = [J] [C]$^{-1}$ = [kg m^2 s^{-2}] [A s]$^{-1}$
- 1 volt = 1 kg m^2 s^{-3} A^{-1} *[1 mark]*

[Total: 2 marks]

> *Recall that potential difference is defined as the energy transferred per unit charge, hence the units 'J C^{-1}'. Parts (b) and (c) of this question can be used to answer part (d). It can save time to look back through previous question parts to see if they can help you.*

Question 4 Medium [11 marks]

Part (a) [6 marks]

Define the following quantities and state their SI base units

(i) velocity

[2]

(ii) acceleration

[2]

(iii) density

[2]

Mark Scheme (6 marks):

(i) Velocity is defined as

- Change in displacement per unit time; *[1 mark]*
- SI base unit = m s^{-1} *[1 mark]*

(ii) Acceleration is defined as

- Change in velocity per unit time; *[1 mark]*
- SI base unit = m s^{-2} *[1 mark]*

(iii) Density is defined as

- Mass per unit volume; *[1 mark]*
- SI base unit = kg m^{-3} *[1 mark]*

[Total: 6 marks]

Part (b) [2 marks]

The pressure P due to a liquid of density ρ is related to the depth h by the expression

$$P = \rho g h$$

where g is the acceleration of free fall.

Use this expression to determine the derived units of pressure P.

Mark Scheme (2 marks):

Use the expression to determine the derived unit for pressure:

- $P = \rho g h$
- Unit of density ρ = kg m^{-3}
- Unit of acceleration g = m s^{-2}
- Unit of height h = m

Determine the unit of pressure:

- $P = \rho g h$
- P = (kg m^{-3}) × (m s^{-2}) × (m) *[1 mark]*
- P = kg m^{-1} s^{-2} *[1 mark]*

[Total: 2 marks]

Part (c) **[3 marks]**

The speed v of a sound wave through a gas of pressure P and density ρ is given by the equation

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

where γ is a constant.

Show that the constant γ has no unit.

Mark Scheme (3 marks):

Show that the constant γ has no unit:

- $v = \sqrt{\frac{\gamma P}{\rho}}$
- Unit of density ρ = kg m^{-3}
- Unit of velocity v = m s^{-1}
- Unit of pressure P = kg m^{-1} s^{-2}

Determine the units on both sides of the expression:

- $v = \sqrt{\frac{\gamma P}{\rho}}$
- LHS: v = m s^{-1} *[1 mark]*
- RHS: $\sqrt{\frac{P}{\rho}} = \sqrt{\frac{kg\ \ m^{-1}\ \ s^{-2}}{kg\ \ m^{-3}}}$ *[1 mark]*
- RHS: $\sqrt{\frac{P}{\rho}} = \sqrt{m^{-1}\ \ m^{3}\ \ s^{-2}} = \sqrt{m^{2}\ \ s^{-2}} = m\ \ s^{-1}$ *[1 mark]*
- LHS = RHS, so γ has no unit

[Total: 3 marks]

Question 5 Medium [4 marks]

Part (a) [2 marks]

Derive the SI base unit of force.

Mark Scheme (2 marks):

Derive the SI base unit of force:

- Force: $F = ma$ *[1 mark]*
- Unit of mass: kg
- Unit of acceleration: m s^{-2}

Determine the unit of force:

- $F = ma$
- F = kg m s^{-2} *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

As a spherical object of radius r moves through the air at speed v, it experiences a force F due to air resistance.

The resistive force F is given by

$$F = krv$$

Where k is a constant.

Derive the SI base unit of the constant k.

Mark Scheme (2 marks):

Derive the SI base unit of the constant k:

- $F = krv$
- Unit of force F = kg m s^{-2}
- Unit of radius r = m
- Unit of speed v = m s^{-1}

Determine the units on both sides of the expression:

- $F = krv \quad \Rightarrow \quad k = \frac{F}{rv}$
- $k = \frac{kg\ m\ s^{-2}}{m\ s^{-1}}$ *[1 mark]*
- k = kg m^{-1} s^{-1} *[1 mark]*

> ***[Total: 2 marks]***

Question 6 Medium [10 marks]

Part (a) [5 marks]

Fig 1.2 shows a turbine that is used to generate electrical power *P* from wind travelling with velocity *v*.

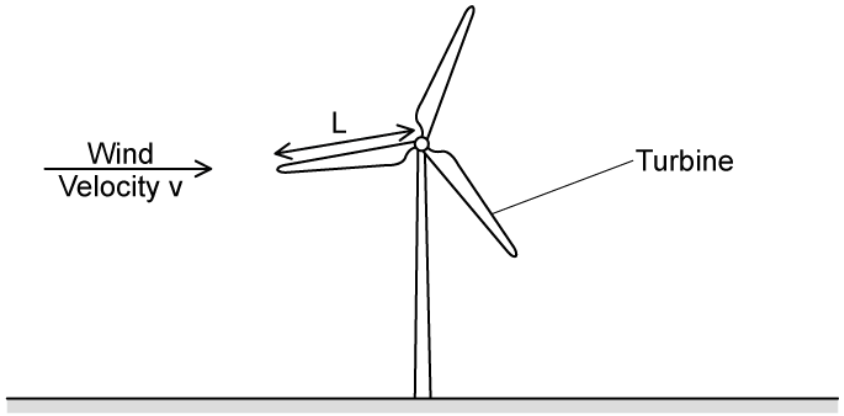


Fig. 1.2

The useful power output *P* of the turbine is given by

$$P = C_P L^2 \rho v^3$$

Where *L* is the length of each blade, ρ is the density of air, *v* is the wind velocity and C_P is a constant.

Show that the constant C_P has no units.

Mark Scheme (5 marks):

Show that the constant C_P has no units:

- $P = C_P L^2 \rho v^3$
- Unit of length L = m
- Unit of density $\rho = \frac{m}{V}$ = kg m^{-3} *[1 mark]*
- Unit of velocity v = m s^{-1}

Determine the SI unit of power:

- Unit of power $P = \frac{E}{t} = \frac{mv^2}{t} = \frac{kg \ m \ s^{-1\,2}}{s}$ *[1 mark]*
- P = kg m^2 s^{-2} s^{-3} = kg m^2 s^{-3} *[1 mark]*

Determine the units on both sides of the expression:

- $P = C_P L^2 \rho v^3$
- LHS: P = kg m^2 s^{-3}

- RHS: $L^2 \rho v^3$ = (m)^2 (kg m^{-3}) (m s^{-1})^3 *[1 mark]*
- RHS: $L^2 \rho v^3$ = kg m^2 m^{-3} m^3 s^{-1} = kg m^2 s^{-3} *[1 mark]*
- LHS = RHS, so C_P has no unit

[Total: 5 marks]

Part (b) [5 marks]

A local council is looking into building wind turbines to supply power to 2500 households in a village. The average energy consumption of a household in the village is 2800 kWh a year.

Fig. 1.3 shows an image from a catalogue detailing the different sizes and power outputs of available wind turbines.

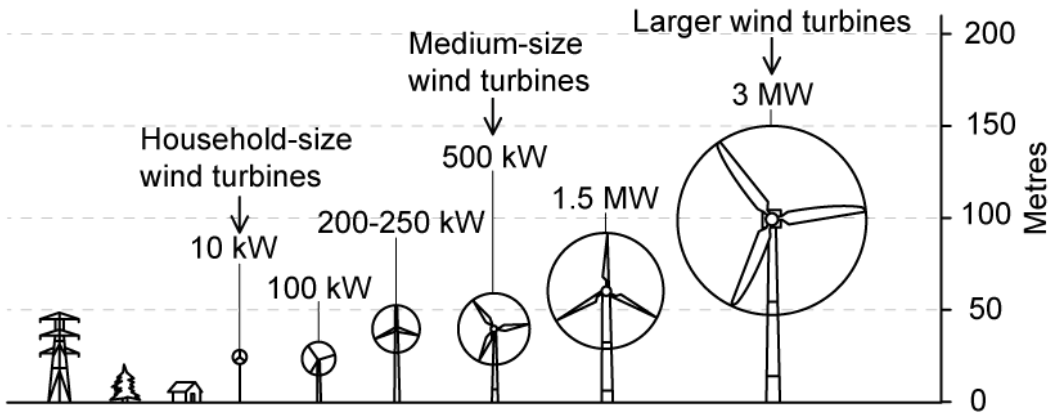


Fig. 1.3

They decide to build 8 identical wind turbines to supply the entire village with electricity.

(i) Determine which of the wind turbines in Fig. 1.3 the council decided to build.

[3]

(ii) The chosen turbine delivers the power specified at a C_P value of 0.45. This value can only be achieved above certain wind velocities.

Estimate the average wind velocity needed to power the village for a year. The density of air is 1.3 kg m^{-3}.

[2]

Mark Scheme (5 marks):

List the known quantities:

- Number of households = 2500
- Average energy consumption per household per year = 2800 kWh

- Number of turbines = 8
- Constant, C_P = 0.45
- Density of air, ρ = 1.3 kg m^{-3}

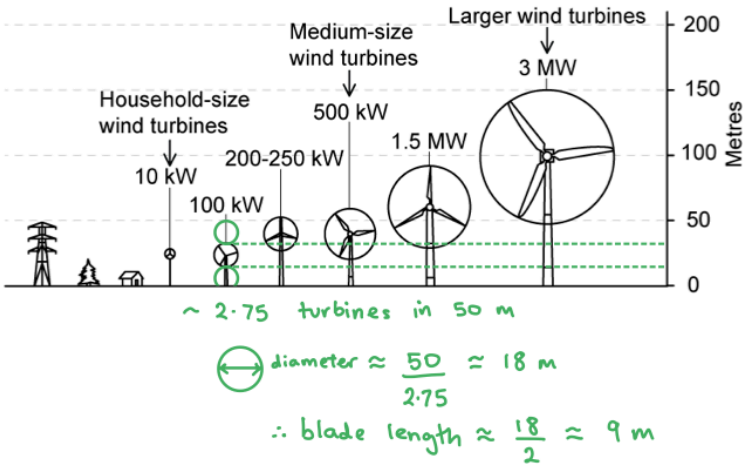
(i) Determine the power rating of the wind turbine the council decided to build:

- Total energy (kWh) required by the village in a year: 2500 × 2800 = 7 000 000 kWh
- Total energy required per day: $\frac{7\ 000\ 000}{365}$ = 19 178 kW h
- Power output required: $\frac{19\ 178}{24}$ = 799 kW *[1 mark]*
- Power output required per turbine: $\frac{799}{8} \approx$ 100 kW *[1 mark]*
- The council chose the 100 kW turbine; *[1 mark]*

(ii) Estimate the average wind velocity required to power the entire village:

- (From Fig 1.3) there are about 2 ¾ 100 kW turbine diameters in 50 m
- So, the diameter of the 1 turbine ≈ $\frac{50}{2.75}$ ≈ 18 m
- The length of a blade L ≈ 9 m *[1 mark]*

  Acceptable range for blade length L = 7 to 11 m



Determine the average wind velocity:

- $P = C_P L^2 \rho v^3 \quad \Rightarrow \quad v = \sqrt[3]{\frac{P}{C_P L^2 \rho}}$
- $v \approx \sqrt[3]{\frac{100\ 000}{0.45 \times 9^2 \times 1.3}} \approx$ 13 m s^{-1} *[1 mark]*
- The wind velocity must average at least 13 m s^{-1} in order for the turbines to maintain a power output of 100 kW

[Total: 5 marks]

Question 7 Hard [13 marks]

Part (a) [6 marks]

Convert the following measurements

(i) 42 km^3 into mm^3

[2]

(ii) 25 light-years into nm

[2]

(iii) 0.05 kWh into PeV

[2]

Mark Scheme (6 marks):

(i) Convert 42 km^3 into mm^3:

- Convert km into mm:
  - 1 km = $\frac{1 \times 10^3}{1 \times 10^{-3}}$ = 1×10^6 mm
- Convert km^3 into mm^3:
  - 1 km^3 = $(1 \times 10^6)^3$ mm^3 = 1×10^{18} mm^3 *[1 mark]*
- 42 km^3 = $42 \times (1 \times 10^{18})$
- 42 km^3 = 4.2×10^{19} mm^3 *[1 mark]*

(ii) Convert 25 light-years into nm:

- 1 light year = the distance light travels in 1 year
- Speed of light = 3.0×10^8 m s^{-1}
- Convert 1 year into s:
  - 1 year = $60 \times 60 \times 24 \times 365 = 3.15 \times 10^7$ s
- Convert 1 ly into m:
  - distance = speed × time
  - 1 ly = $(3.0 \times 10^8) \times (3.15 \times 10^7) = 9.46 \times 10^{15}$ m *[1 mark]*
- Convert m into nm:
  - 1 ly = $\frac{9.46 \times 10^{15}}{1 \times 10^{-9}}$ = 9.46×10^{24} nm
- 25 ly = $25 \times (9.46 \times 10^{24})$
- 25 ly = 2.4×10^{26} nm *[1 mark]*

(iii) Convert 0.05 kWh into PeV:

- Convert from kW h to W h:
  - 1 kW = 1000 W h
  - 0.05 kWh = 0.05 × 1000 = 50 W h
- Convert from W h to J:
  - 1 h = 60 × 60 = 3600 s
  - 50 W h = 50 × 3600 = 180 000 W s = 180 000 J
- Convert J to eV:
  - 1 eV = 1.60×10^{-19} J
  - 180 000 J = $\frac{180\ 000}{1.60 \times 10^{-19}}$ = 1.125×10^{24} eV *[1 mark]*
- Convert from eV to PeV:
  - 1 PeV = 1×10^{15} eV
- 0.05 kWh = $\frac{1.125 \times 10^{24}}{1 \times 10^{15}}$
- 0.05 kWh = 1.1×10^{9} PeV *[1 mark]*

[Total: 6 marks]

Part (b) **[3 marks]**

The frequency *f* of a stationary wave on a string is given by

$$f^{k} = \frac{T}{4mL^{2}}$$

where *L* is the length of the string, *T* is the tension in the string, *m* is the mass per unit length of the string and *k* is a dimensionless constant.

By considering the homogeneity of the equation, determine the value of *k*.

Mark Scheme (3 marks):

By considering the homogeneity of the equation, determine the value of *k*:

- $f^{k} = \frac{T}{4mL^{2}}$
- Consider the units on the LHS: $\mathrm{Hz}^{k} = \mathrm{s}^{-1\,k} = \mathrm{s}^{-k}$ *[1 mark]*
- Consider the units on the RHS: $\frac{\mathrm{N}}{\mathrm{kg\ m^{-1}\ m^{2}}} = \frac{\mathrm{N}}{\mathrm{kg\ m}}$ *[1 mark]*

Simplify the units on the RHS:

- From $F = ma$, the SI base unit of N is $\mathrm{kg\ m\ s^{-2}}$
- Hence $\frac{\mathrm{N}}{\mathrm{kg\ m^{-1}\ m^{2}}} = \frac{\mathrm{kg\ m\ s^{-2}}}{\mathrm{kg\ m}} = \mathrm{s^{-2}}$

- Comparing RHS with LHS: $s^{-k} = s^{-2}$
- Therefore, $k = 2$ *[1 mark]*

[Total: 3 marks]

Part (c) **[4 marks]**

The current I through a semiconductor diode given by

$$I = I_0 \ e^{-\frac{qV}{nT}}$$

where I_0 is the initial current, q is the charge of an electron, V is the applied potential difference, T is the temperature and n is a constant.

(i) By considering the homogeneity of the equation, determine the units of n.

[3]

(ii) Use the datasheet to determine the name of this constant.

[1]

Mark Scheme (4 marks):

(i) By considering the homogeneity of the equation, determine the units of n:

- $I = I_0 \ e^{-\frac{qV}{nT}} \quad \Rightarrow \quad \frac{I}{I_0} = e^{-\frac{qV}{nT}}$
- Since the units on the LHS $\frac{A}{A} = 1$, the units of $\frac{qV}{nT}$ will also be dimensionless

State the SI units of temperature T:

- Units of $T = \text{K}$

Determine the SI units of charge q:

- Charge: $I = \frac{q}{t} \quad \Rightarrow \quad q = It$
- Units of $q = \text{C} = \text{A s}$ *[1 mark]*

Determine the SI units of potential difference V:

- Potential difference: $V = \frac{W}{q}$
- Work done: $W = F \times s$ or $\frac{1}{2}mv^2$
- Units of $W = mv^2 = \text{J} = \text{kg m}^2\text{ s}^{-2}$
- Units of $V = \text{J C}^{-1} = \text{J A}^{-1}\text{ s}^{-1}$
- Units of $V = \text{kg m}^2\text{ s}^{-2}\text{ s}^{-1}\text{ A}^{-1} = \text{kg m}^2\text{ s}^{-3}\text{ A}^{-1}$ *[1 mark]*

Determine the units of constant n:

- Units of n: $n = \frac{qV}{T}$

- $n = \frac{\text{A s kg m}^2\ \text{s}^{-3}\ \text{A}^{-1}}{\text{K}}$
- $n = \text{kg m}^2\ \text{s}^{-2}\ \text{K}^{-1}$ *[1 mark]*

(ii) Determine the name of the constant:

- Since $\text{J} = \text{kg m}^2\ \text{s}^{-2}$, the derived unit of *n* is J K^{-1}
- Therefore, this is the Boltzmann constant, k = 1.38 × 10^{-23} J K^{-1} *[1 mark]*

[Total: 4 marks]

Question 8 Hard [12 marks]

Part (a) [2 marks]

The diameter of a proton is approximately 1 fm.

Estimate the time it takes light to cross the nucleus of a hydrogen atom.

1 fm = 1×10^{-15} m

Mark Scheme (2 marks):

Estimate the time it takes light to cross the nucleus of a hydrogen atom:

- Speed of light in vacuum, $c = 3.00 \times 10^{8}$ m s^{-1}

Estimate the diameter of the nucleus of a hydrogen atom:

- A hydrogen atom has a nucleus made of 1 proton
- The size of a hydrogen nucleus is 1 fm = 1×10^{-15} m

Estimate the time using the equation relating speed, time and distance:

- $\text{time} = \frac{\text{distance}}{\text{speed}}$
- $\text{time} = \frac{1 \times 10^{-15}}{3.00 \times 10^{8}}$ *[1 mark]*
- $\text{time} = 3 \times 10^{-24}$ s *[1 mark]*

[Total: 2 marks]

Part (b) [4 marks]

A foil made from gold has a thickness of 0.132 μm. A gold atom has a radius of 146 pm.

(i) Estimate the number of layers of atoms in the gold foil to the nearest hundred.

[1]

(ii) The radius of a gold nucleus is about 20 000 times smaller than its atomic radius. The cross-sectional area of a nucleus is commonly measured using the unit *barn*, which is represented by the symbol *b*.

Estimate the cross-sectional area of a gold nucleus in barns (b).

1 b = 100 fm^2

[3]

Mark Scheme (4 marks):

List the known quantities:

- Thickness of gold foil = 0.132 μm = 0.132×10^{-6} m

- Radius of gold atom = 146 pm = 146 × 10^{-12} m
- Radius of gold nucleus = $\frac{\text{radius of gold atom}}{20\ 000}$

(i) Estimate the number of layers of atoms in the gold foil to the nearest hundred:

- Diameter of a gold atom = (146 × 10^{-12}) × 2 = 292 × 10^{-12} m
- Number of layers = $\frac{0.132 \times 10^{-6}}{292 \times 10^{-12}}$ = 452.05 = 500 layers (1 s.f.) *[1 mark]*

(ii) Calculate the cross-sectional area of a gold nucleus in barns (b):

- Convert fm^2 into m^2:
  - 1 fm = 1 × 10^{-15} m
  - 1 fm^2 = (1 × 10^{-15})^2 m^2 = 1 × 10^{-30} m^2
- Calculate the value of 1 barn in m^2:
  - 1 b = 100 fm^2 = 100 × (1 × 10^{-30}) m^2 = 1 × 10^{-28} m^2 *[1 mark]*

Determine the cross-sectional area and convert to barns:

- Radius of a gold nucleus: $r \approx \frac{146 \times 10^{-12}}{20\ 000} \approx 7.3 \times 10^{-15}$ m
- Cross-sectional area of a gold nucleus = $\pi r^2 \approx \pi 7.3 \times 10^{-15\ 2} \approx 1.67 \times 10^{-28}$ m^2 *[1 mark]*
- Cross-sectional area of a gold nucleus = $\frac{1.67 \times 10^{-28}}{1 \times 10^{-28}} \approx 1.67$ b *[1 mark]*

[Total: 4 marks]

Although the values in the question are given to three significant figures, it makes sense to round to the nearest hundred in part (i), since we are ignoring factors such as spaces between atoms and layers in the lattice.

The barn described in this question is a real unit. Other less well-known units include the 'outhouse' (1 μb) and the 'shed' (10^{-24} b). Particle physicists are clearly quite creative!

Part (c) **[3 marks]**

The atomic mass unit *u* is a common unit used in nuclear physics. 1*u* is equivalent to 931.5 MeV c^{-2}

Show that 1*u* is approximately equal to the mass of a proton in kg.

Mark Scheme (3 marks):

Show that 1*u* is approximately equal to the mass of a proton in kg:

List the known quantities:

- Atomic mass unit, 1 *u* = 931.5 MeV c^{-2}
- Speed of light in vacuum, c = 3.00 × 10^8 m s^{-1}
- Mass of a proton, m_p = 1.67 × 10^{-27} kg

Convert 1u to J:

- 1u = 931.5 MeV = 931.5 × 10^6 eV *[1 mark]*
- 1 eV = 1.60 × 10^{-19} J
- 1u = (931.5 × 10^6) × (1.60 × 10^{-19}) = 1.4904 × 10^{-10} J *[1 mark]*

Convert energy in J to mass in kg:

- Energy-mass equivalence: $\Delta E = \Delta m c^2 \Rightarrow \Delta m = \frac{\Delta E}{c^2}$
- $m = \frac{1.4904 \times 10^{-10}}{(3.00 \times 10^8)^2}$ = 1.66 × 10^{-27} kg *[1 mark]*
- Therefore, 1u is approximately equal to the mass of a proton m_p = 1.67 × 10^{-27} kg

[Total: 3 marks]

You will be expected to remember the definition of an electronvolt 1 eV = 1.60 × 10^{-19} J for your exam, however, you will be given the charge on an electron e on your data sheet which is also equal to 1.60 × 10^{-19} C

Part (d) [3 marks]

An acre is a unit of land measurement approximately equal to 4050 m^2, which is roughly equivalent to the size of a football field.

An Astronomical Unit, or 1 AU, is defined as the average distance between the Earth and the Sun and is approximately equal to 150 million km.

Estimate the number of football fields that could fit lengthwise between the Earth and the Sun.

Mark Scheme (3 marks):

Estimate the number of football fields that could fit lengthwise between the Earth and the Sun:

- 1 AU = 150 million km = 150 × 10^6 × 10^3 m = 150 × 10^9 m *[1 mark]*
- 1 acre = 1 football field = 4050 m^2

Estimate the length of a football field

- A football field is rectangular, so a reasonable ratio of the length to the width of a football field might be 3:2
- This means area: $A = L \times w \Rightarrow A = 3x \times 2x$
- Where x = the constant of proportionality in the 3:2 ratio, i.e. $L = 3x$ and $w = 2x$
- $4050 = 6x^2 \Rightarrow x = \sqrt{\frac{4050}{6}} = 25.981 \approx 26$ m
- Hence, length: $L = 3x \approx 3 \times 26 \approx 78$ m *[1 mark]*

Estimate the number of lengths that would fit in 1 AU:

- Number of football fields in 1 AU = $\frac{150 \times 10^9}{78}$ = 1.9231 × 10^9 ≈ 2 × 10^9 (1 s.f.) *[1 mark]*

- Therefore, approximately 2 billion football fields could fit lengthwise between the Earth and the Sun

[Total: 3 marks]

Questions such as this are all about your problem-solving skills. While approximating a football field to a square might not be exactly right, if you took this approach, then you would still end up with the same answer once you round down to 1 significant figure

Approximating the field as a square: length = $\sqrt{4050}$ ≈ 64 m

Number of football fields in 1 AU = $\frac{150 \times 10^9}{64}$ = 2.3438 × 10^9 ≈ 2 × 10^9 (1 s.f.)